Component	Pin Function	ESP32 GPIO	Pin Type	Hardware Action / Notes
All ToF Sensors	I2C SDA	GPIO 21	Safe	Connect all 3 sensor SDA pins here.
All ToF Sensors	I2C SCL	GPIO 22	Safe	Connect all 3 sensor SCL pins here.
Front ToF	XSHUT	GPIO 14	Safe	Pull HIGH in code to activate.
Right ToF	XSHUT	GPIO 27	Safe	Pull HIGH in code to activate.
Servo 1	PWM Signal	GPIO 13	Safe	Assign to ESP32 PWM Channel in code.
Servo 2	PWM Signal	GPIO 23	Safe	Assign to ESP32 PWM Channel in code.
Servo 3	PWM Signal	GPIO 32	Safe	Assign to ESP32 PWM Channel in code.
Servo 4	PWM Signal	GPIO 18	Safe	Assign to ESP32 PWM Channel in code.
Servo 5	PWM Signal	**GPIO 2**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Down resistor to GND.**
DC Motor 1	Speed (PWM)	GPIO 33	Safe	Connect to Motor Driver Enable pin.
DC Motor 1	Direction (DIR)	GPIO 19	Safe	Connect to IN1 (and NOT gate to IN2).
DC Motor 2	Speed (PWM)	GPIO 4	Safe	Connect to Motor Driver Enable pin.
DC Motor 2	Direction (DIR)	GPIO 26	Safe	Connect to IN1 (and NOT gate to IN2).
DC Motor 3	Speed (PWM)	GPIO 16	Safe	Connect to Motor Driver Enable pin.
DC Motor 3	Direction (DIR)	**GPIO 5**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Up resistor to 3.3V.**
DC Motor 4	Speed (PWM)	GPIO 17	Safe	Connect to Motor Driver Enable pin.
DC Motor 4	Direction (DIR)	**GPIO 15**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Up resistor to 3.3V.**